LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	21
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Position control servomechanism, speed control servomechanism
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Position control servomechanism, speed control servomechanism.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagram and signal flow; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an AFCS?
(b) Define a servomechanism in the context of aircraft control.
(c) List two types of servomechanisms covered in this lesson.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic operation of position control servomechanism • Components of position control servomechanism • Function of position control servomechanism in AFCS • Error signal generation in position control • Stability considerations in position control	L & Ds	PPT, IB	15 min
(b)	• Definition and basic operation of speed control servomechanism • Components of speed control servomechanism • Function of speed control servomechanism in AFCS • Error signal generation in speed control • Comparison between position and speed control	L & Ds	PPT, PDF	20 min
(c)	• Integration of position and speed control in AFCS • Typical applications in aircraft systems • Troubleshooting common failures • Maintenance checks for servomechanisms • Safety protocols during AFCS operation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the error signal is generated in a position control servomechanism.
(b) Differentiate between the functions of position and speed control servomechanisms in an AFCS.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of position and speed control servomechanisms apply to the response and damping concepts discussed in LP 22.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Response of servomechanism and damping and its types.	Refer to the master syllabus reference list	LP 22

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min